

Intervention Prompt Template

Task: "You are an expert in simulating realistic errors in multi-agent code generation systems. Please modify the internal output of the system to introduce realistic errors that could occur during multi-agent collaboration. Output only the modified feature value, without explanation."

Problem Context:

{problem_description}

Intermediate Output:

{intermediate_outputs}

Instructions:

1. Ensure modifications are internally consistent with each other.
2. Create realistic errors that an AI agent might make (e.g., misunderstanding requirements, wrong assumptions).

Modification Guidelines:

- For "requirement_analysis" fields, introduce ambiguities or misinterpretations of requirements.
- Change "Language" to a different language (e.g., 'zh_cn'), keep other field of output still in English.
- Change "Programming Language" to another language (e.g., 'Java'), keep the other field of code still in python.
- Modify "Original Requirements" to introduce logic changes or different constraints.
- Alter "Product Goals" to shift focus or priorities.
- Change "User Stories" to reflect different user personas or use cases.
- Modify competitive analysis to include different competitors or change positioning.
- Adjust "Requirement Pool" priorities or add conflicting requirements.
- Introduce ambiguity in "Anything UNCLEAR" section.
- Change the implementation approach to use different algorithms or data structures.
- Modify file structure or add/remove files.
- Alter data structures and interfaces (e.g., change function signatures, add parameters).
- Change the program call flow or add new interaction patterns.
- Introduce performance constraints or optimization requirements.
- Add or remove dependencies and third-party libraries.
- Modify design patterns or architectural decisions.
- Change required packages to different versions or alternatives.
- Modify required third-party packages.
- Alter the logic analysis to introduce different implementation steps.
- Change file name in file list or reorganize code structure.
- Modify API specifications.
- Introduce new shared knowledge or constraints.
- Add complexity or simplify implementation details.

Figure 1: Intervention prompt template for counterfactual interventions.